

Lecture 6: Probabilistic Optimization

BINF 4002

Today

1. Questions on problem set or updated course policies
2. Colab notebooks will be added to Courseworks
3. Probabilistic optimization

Yesterday

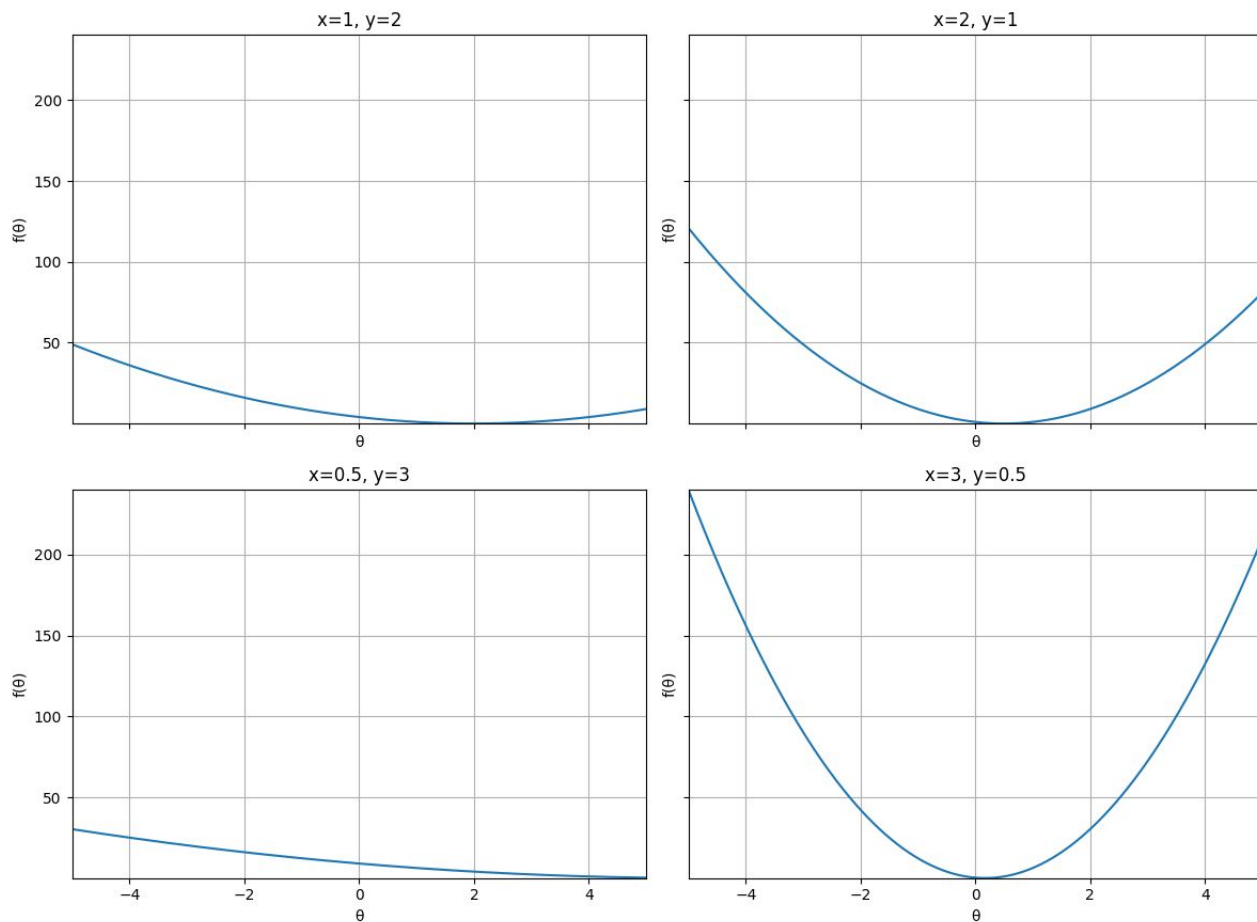
Let

$$f(\theta; y, x) = (y - \theta x)^2$$

What value of θ minimizes $f(\theta; y, x)$?

For scalars,
every value of
 x and y yield a
different
optimization
problem.

Plot of $f(\theta) = (y - \theta x)^2$



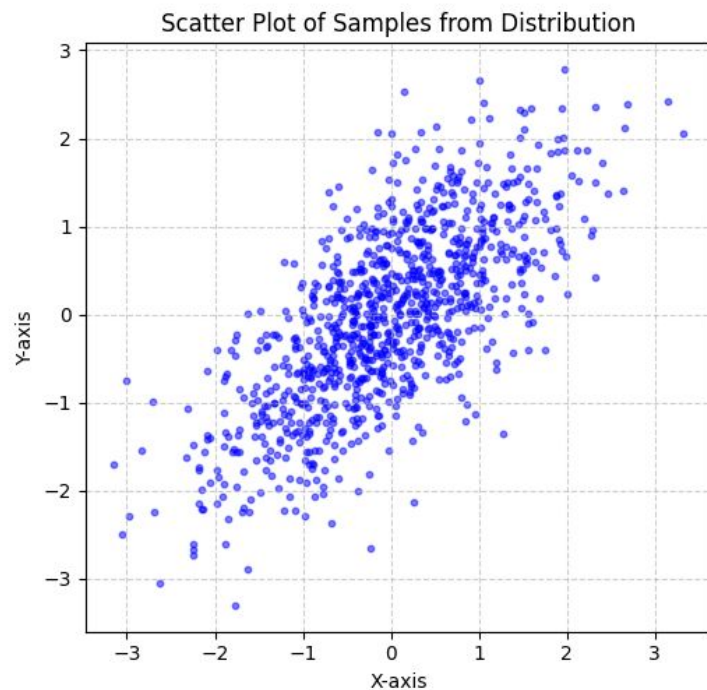
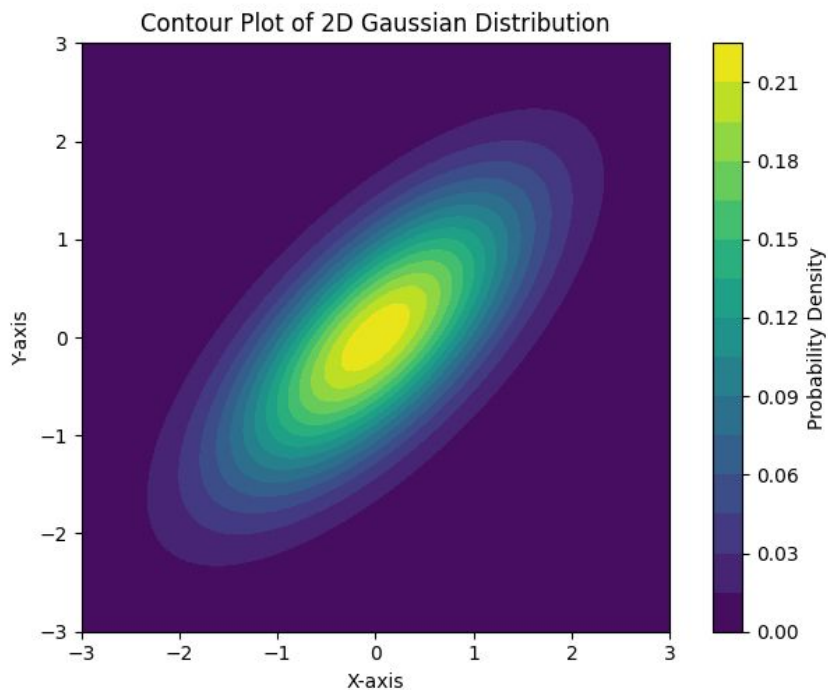
Today

Let $\mathbf{x}, \mathbf{y}$ be random variables described by joint density function $p_{\mathbf{x}, \mathbf{y}}(x, y)$, and let

$$f(\theta; \mathbf{y}, \mathbf{x}) = ???$$

What value of θ minimizes this function?

Now, x and y are sampled from some joint, and we care about their overall relationship.



What do we care about optimizing?

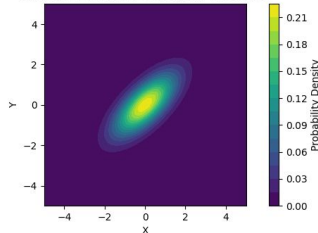
$$f(\theta; y, \mathbf{x}) = ???$$

One natural choice: The Expectation of our old optimization function!

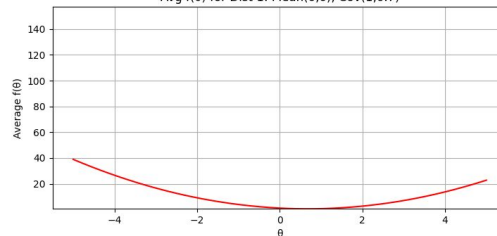
$$f(\theta; y, \mathbf{x}) = \mathbb{E}_{\mathbf{x}, y}[(y - \theta x)^2]$$

Now, for each distribution over x, y , we have a different optimization function!

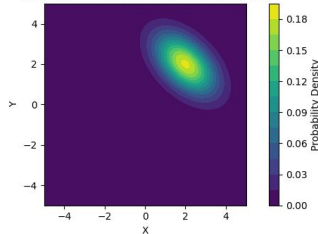
Contour Plot: Dist 1: Mean(0,0), Cov(1,0.7)



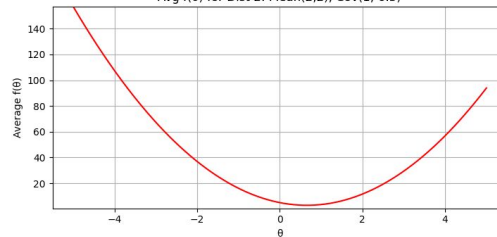
Avg $f(\theta)$ for Dist 1: Mean(0,0), Cov(1,0.7)



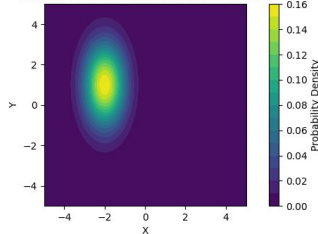
Contour Plot: Dist 2: Mean(2,2), Cov(1,-0.5)



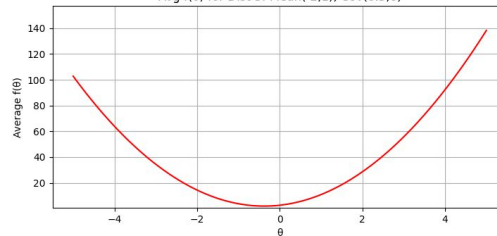
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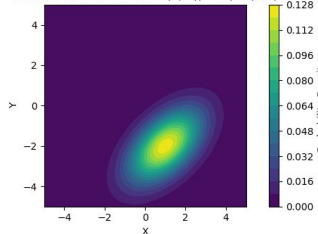
Contour Plot: Dist 3: Mean(-2,1), Cov(0.5,0)



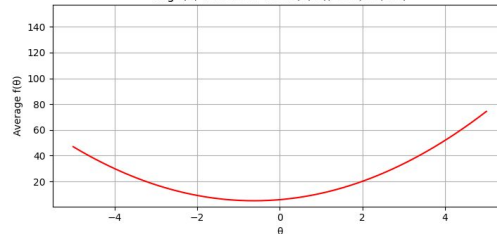
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Contour Plot: Dist 4: Mean(1,-2), Cov(1.5,0.8)



Avg $f(\theta)$ for Dist 4: Mean(1,-2), Cov(1.5,0.8)



Key Differences

Deterministic Optimization

1. Our function is static.
If it is dependent on data, then that data is assumed to be static, and we don't need to worry about the function being evaluated over new, unseen data.

Probabilistic Optimization

1. ???

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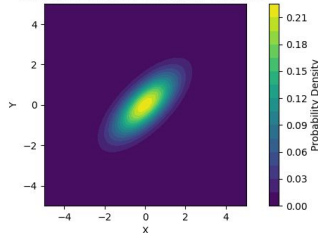
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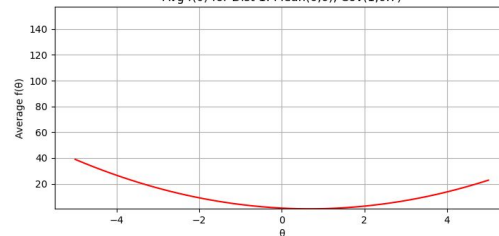
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Now, for each distribution over x, y , we have a different optimization function!

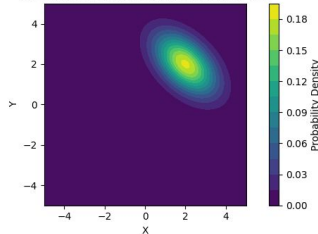
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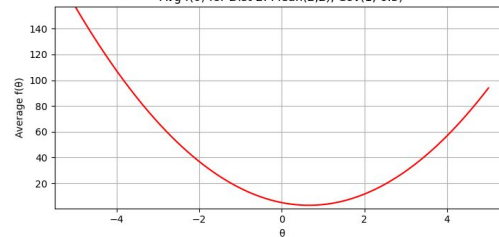
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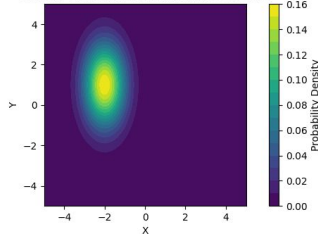
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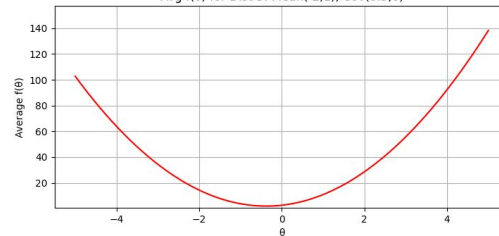
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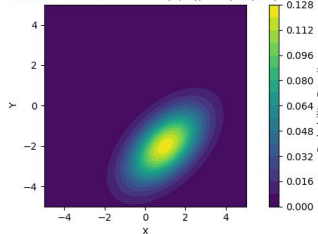
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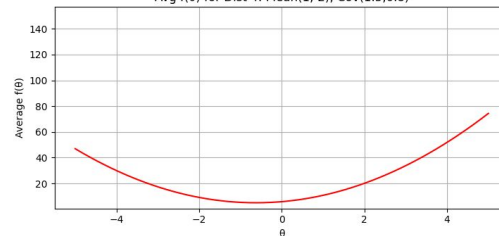
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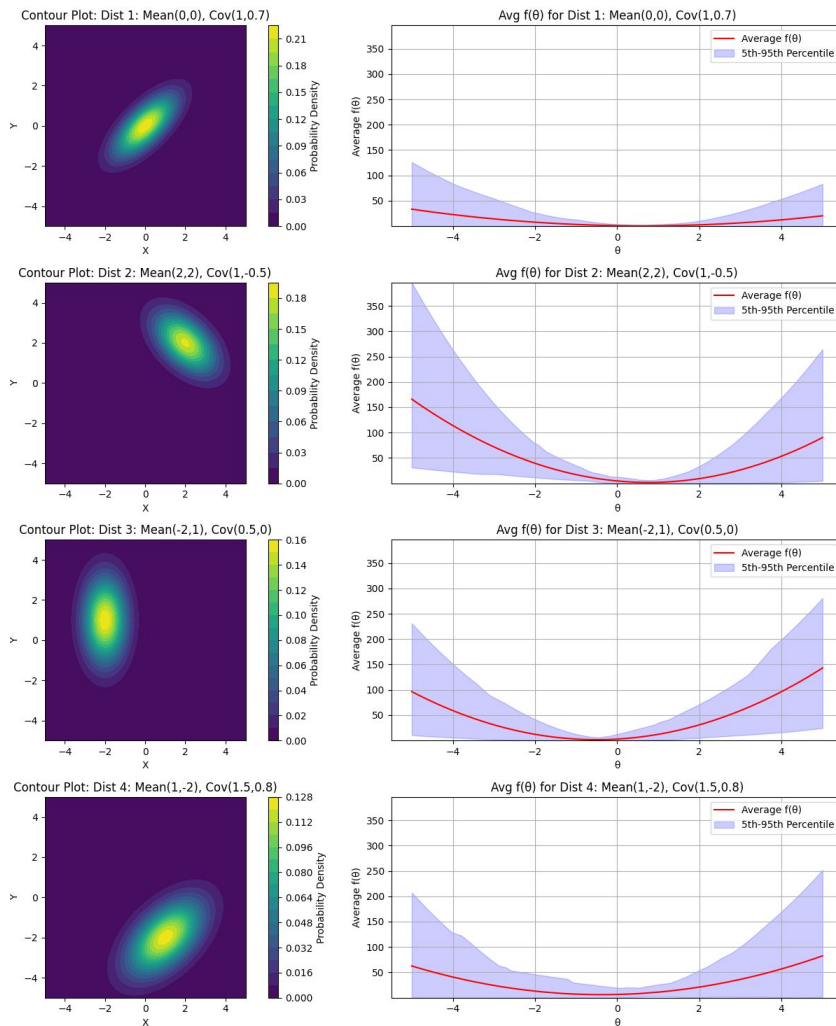
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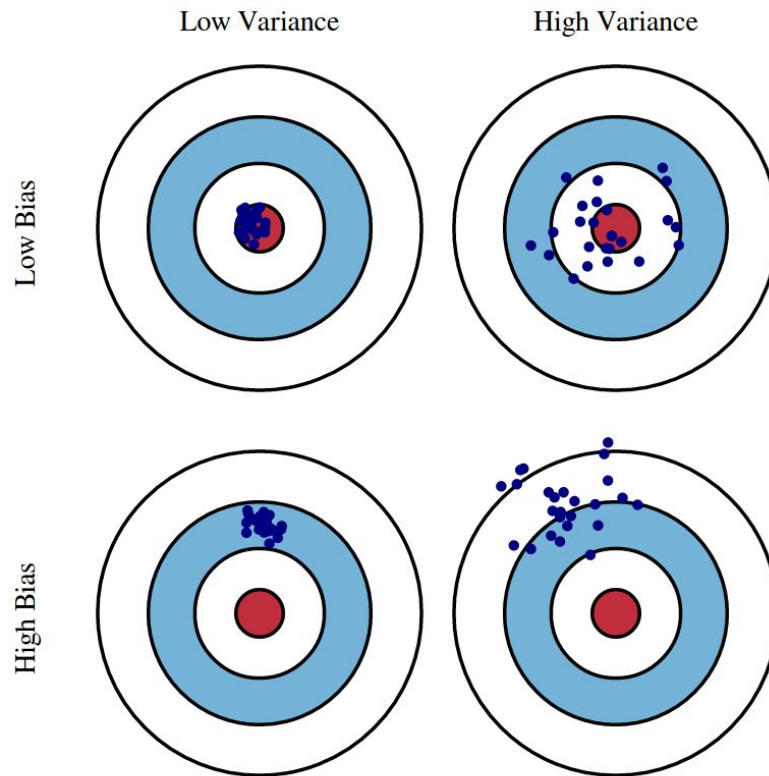
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$$f(\theta; y, \mathbf{x}) = \mathbb{E}_{\mathbf{x}, y}[(y - \theta x)^2]$$

Error bars capture 90%
CI around values of this!



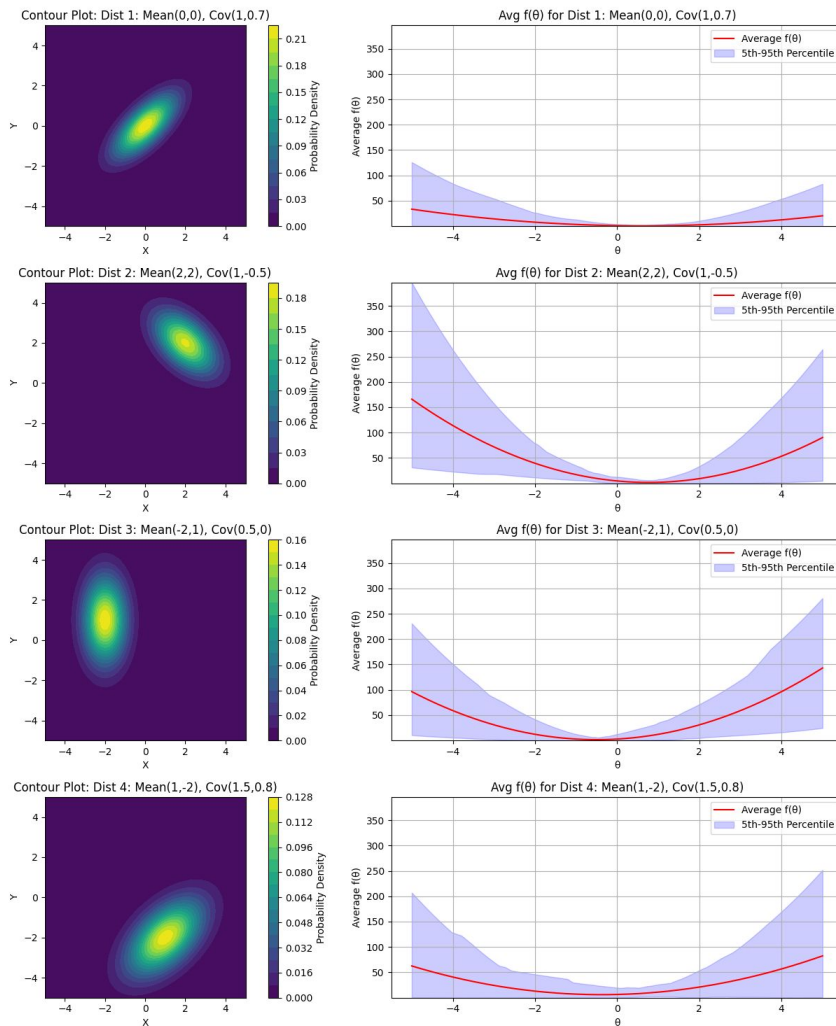
Bias & Variance: Two Sources of Error



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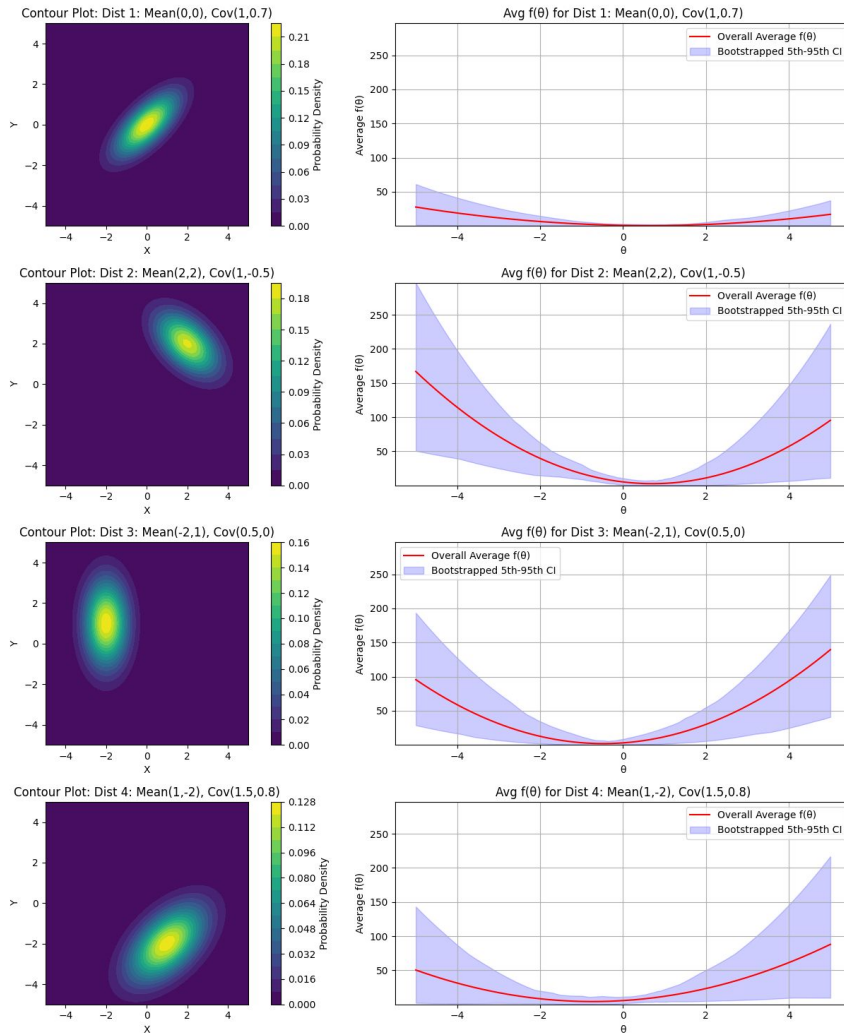
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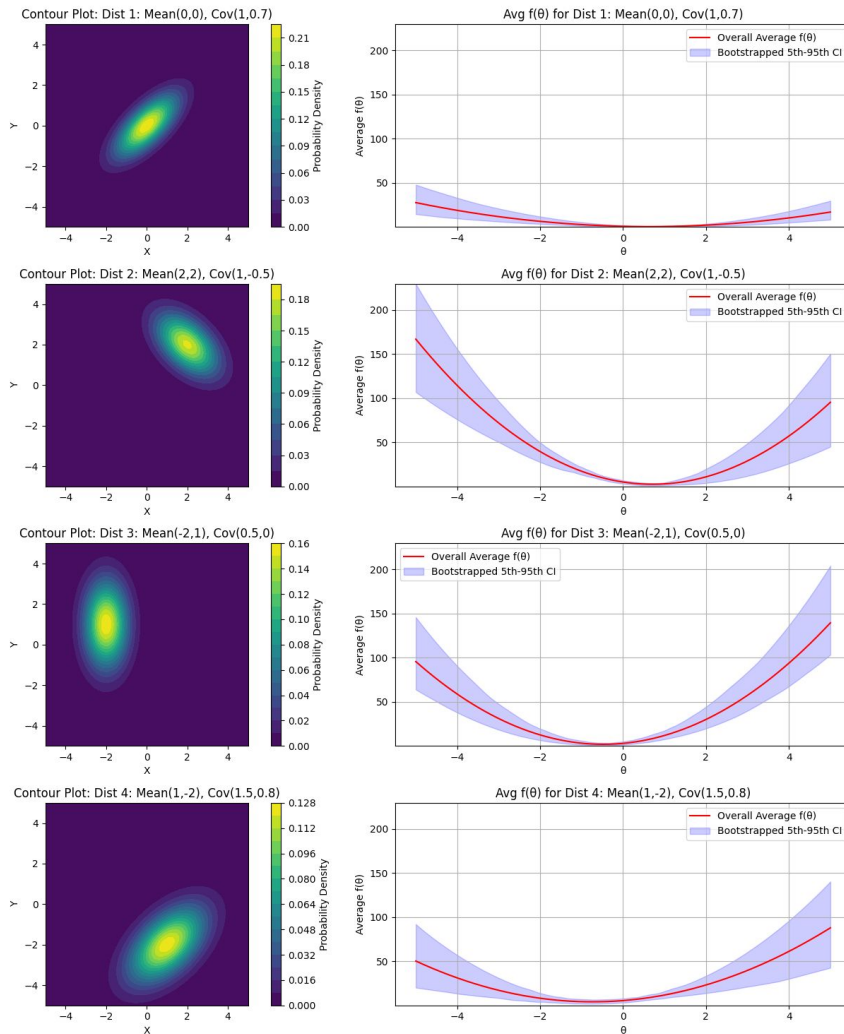
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3. Our algorithms inherently must rely on some degree of stochasticity simply to reflect the problem.

So how do we solve
probabilistic optimization
problems?

Stochastic gradient descent!

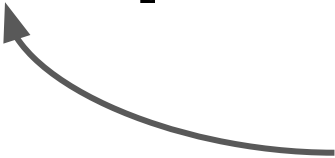
If $f(\theta; \mathbf{y}, \mathbf{x}) = \mathbb{E}_{\mathbf{x}, \mathbf{y}}[(y - \theta x)^2]$, then we can optimize by finding the gradient and descending!

$$\begin{aligned}\nabla_{\theta} f(\theta; \mathbf{y}, \mathbf{x}) &= \frac{\partial}{\partial \theta} \mathbb{E}_{\mathbf{x}, \mathbf{y}}[(y - \theta x)^2] \\ &= \mathbb{E}_{\mathbf{x}, \mathbf{y}} \left[\frac{\partial}{\partial \theta} (y - \theta x)^2 \right]\end{aligned}$$

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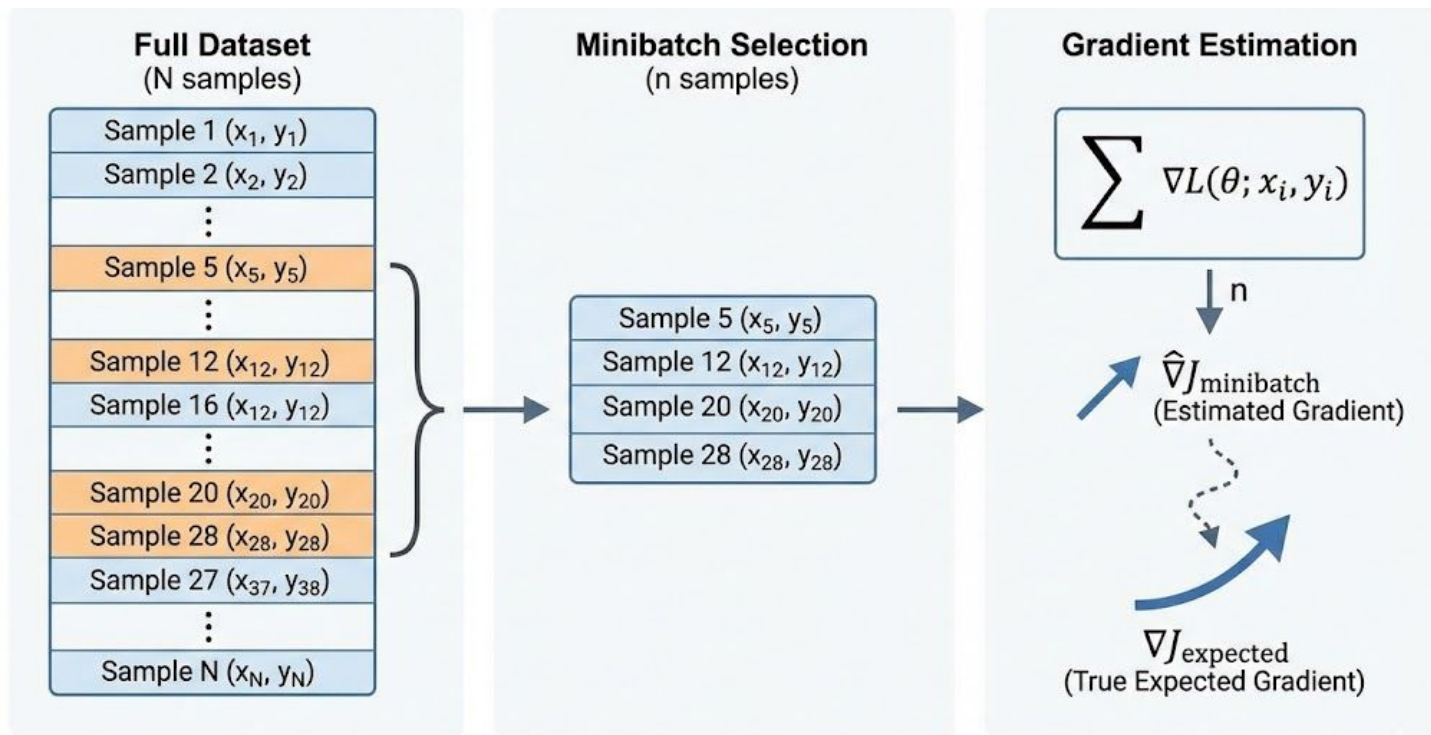
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We can now take a stochastic estimate of this gradient, and descend via that!

Is what I just described
what we do with
minibatch stochastic
gradient descent in AI?

Minibatch SGD

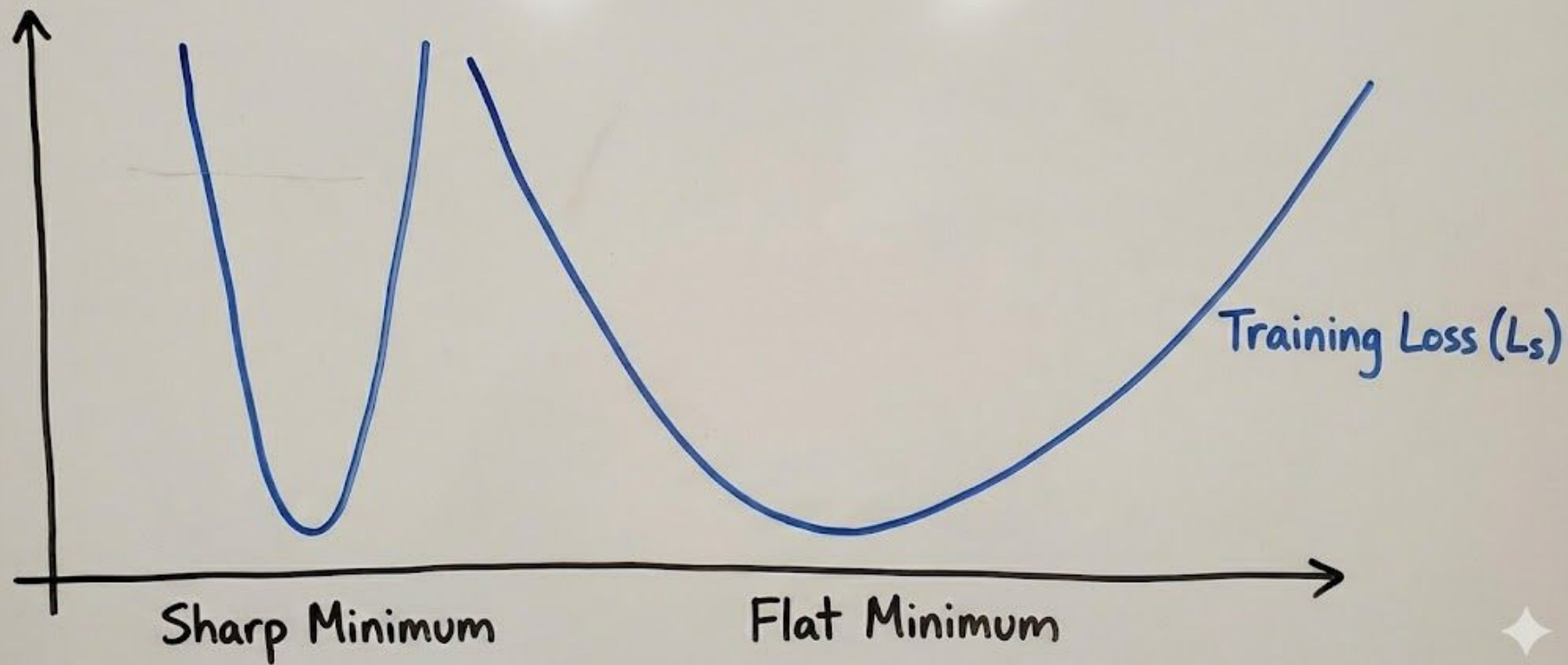


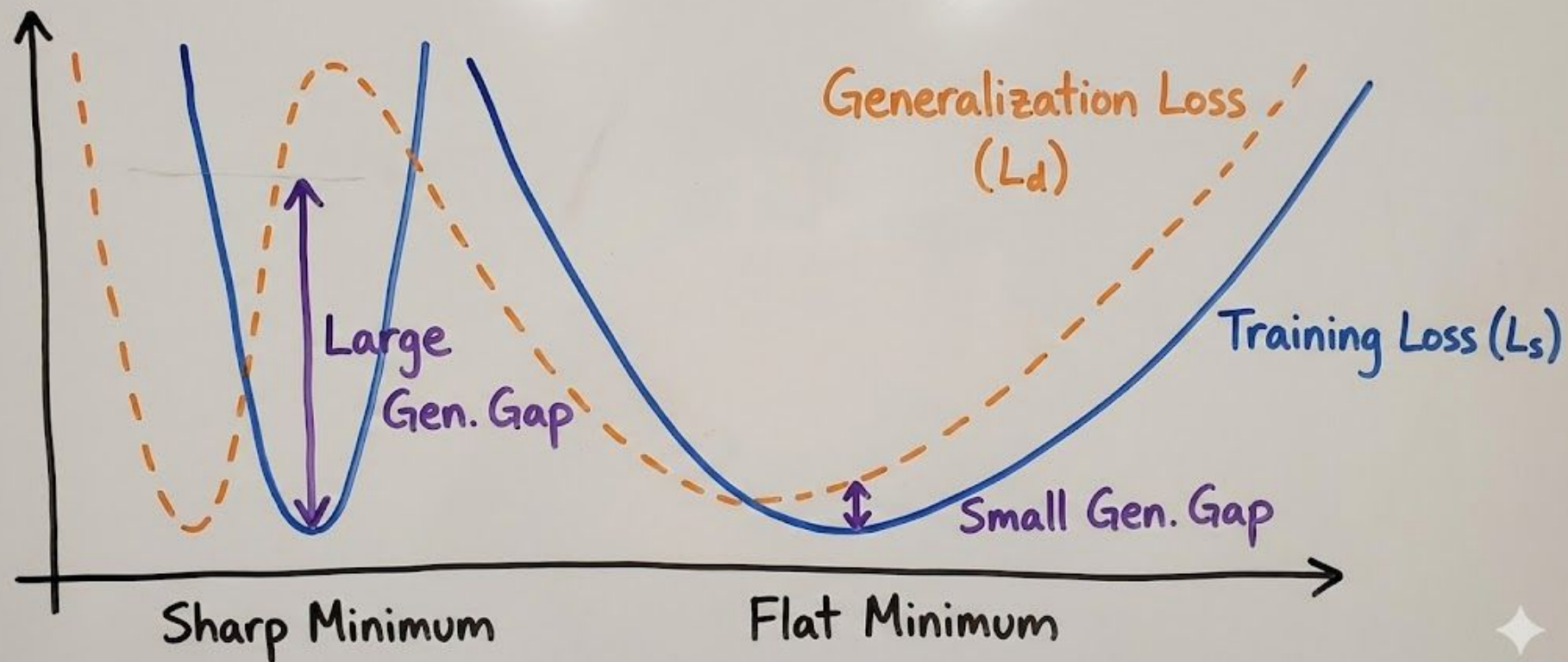
Minibatch SGD estimates the expected gradient using a random subset of data.

2 Concepts in Minibatch SGD:

1. Stochastic Estimate of the Expectation of the Gradient
2. Compute the estimated gradient more cheaply at the expense of increased variance.

Is variance in the
gradient actually bad?
Intro to probabilistic
optimization geometry!





3 Concepts in Minibatch SGD:

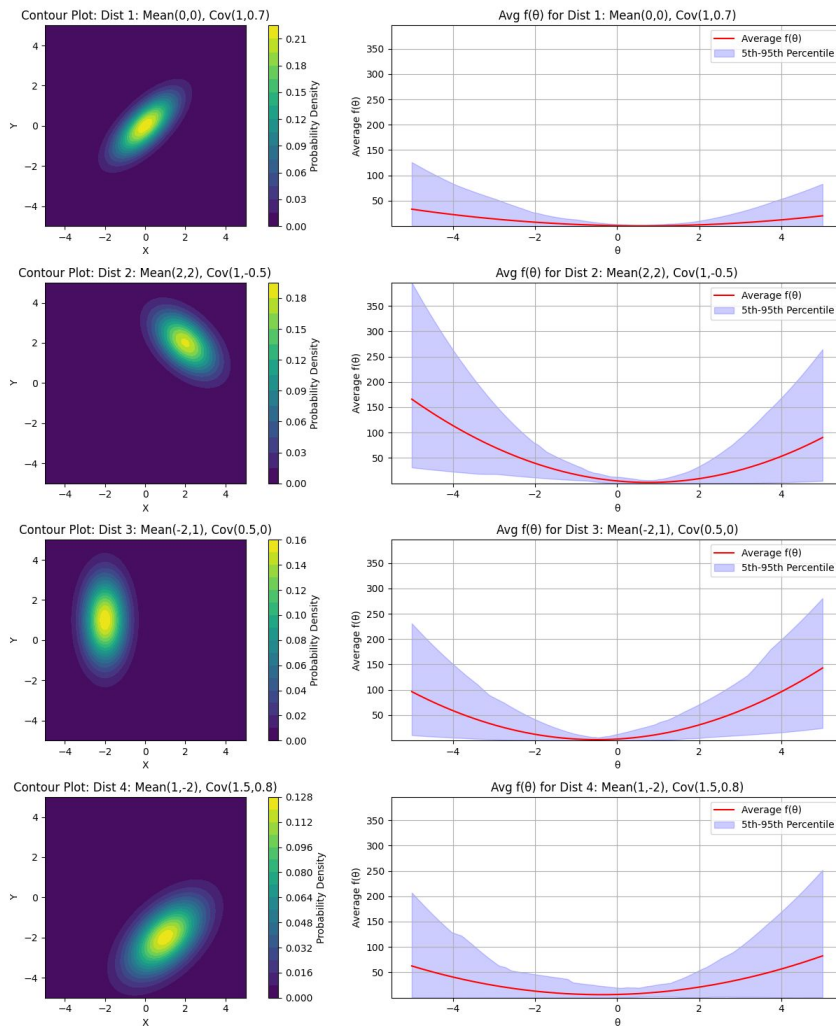
1. Stochastic Estimate of the Expectation of the Gradient
2. Compute the estimated gradient more cheaply at the expense of increased variance.
3. Increase generalizability by increasing variance in sampled gradients!

Generalization vs.
Memorization,
probabilistically

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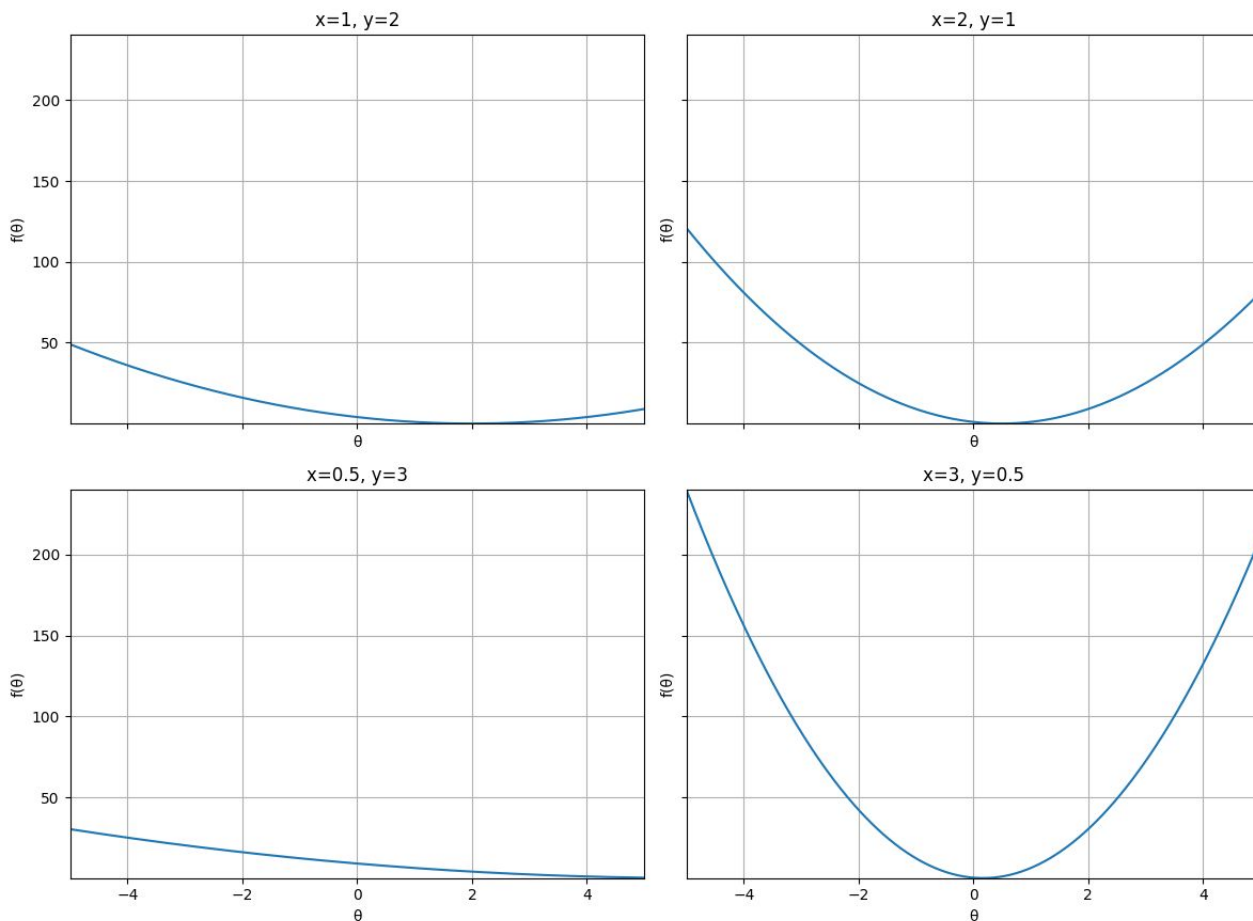
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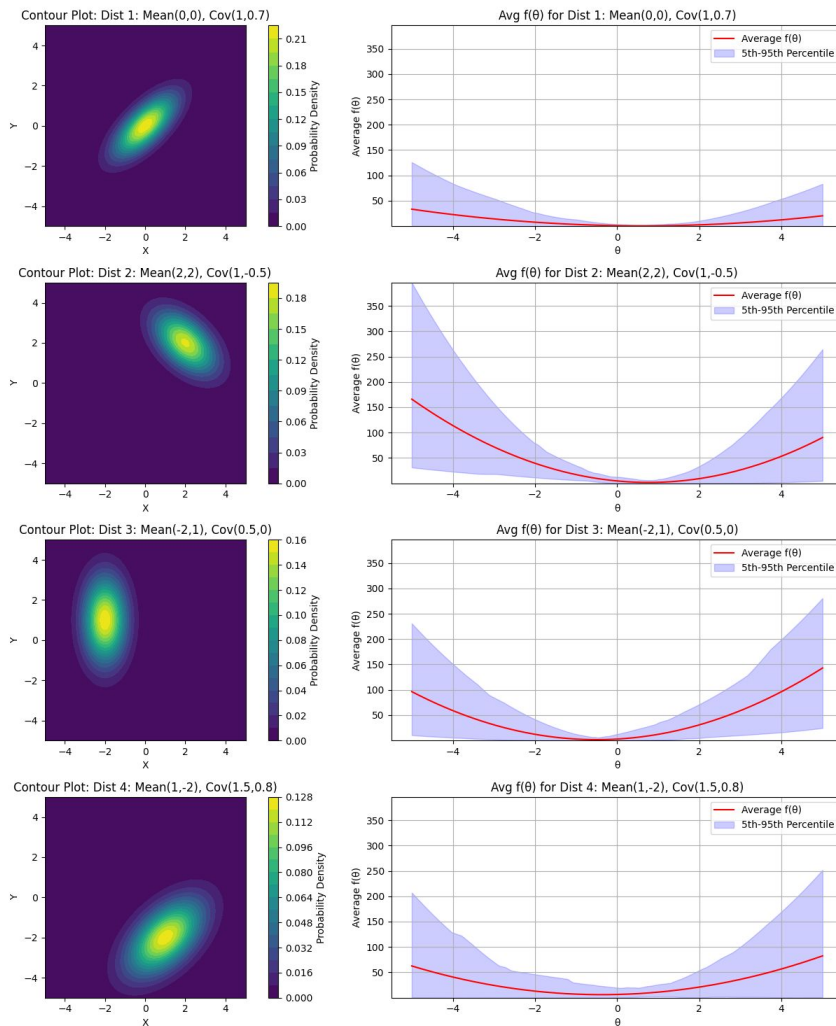
Plot of $f(\theta) = (y - \theta x)^2$



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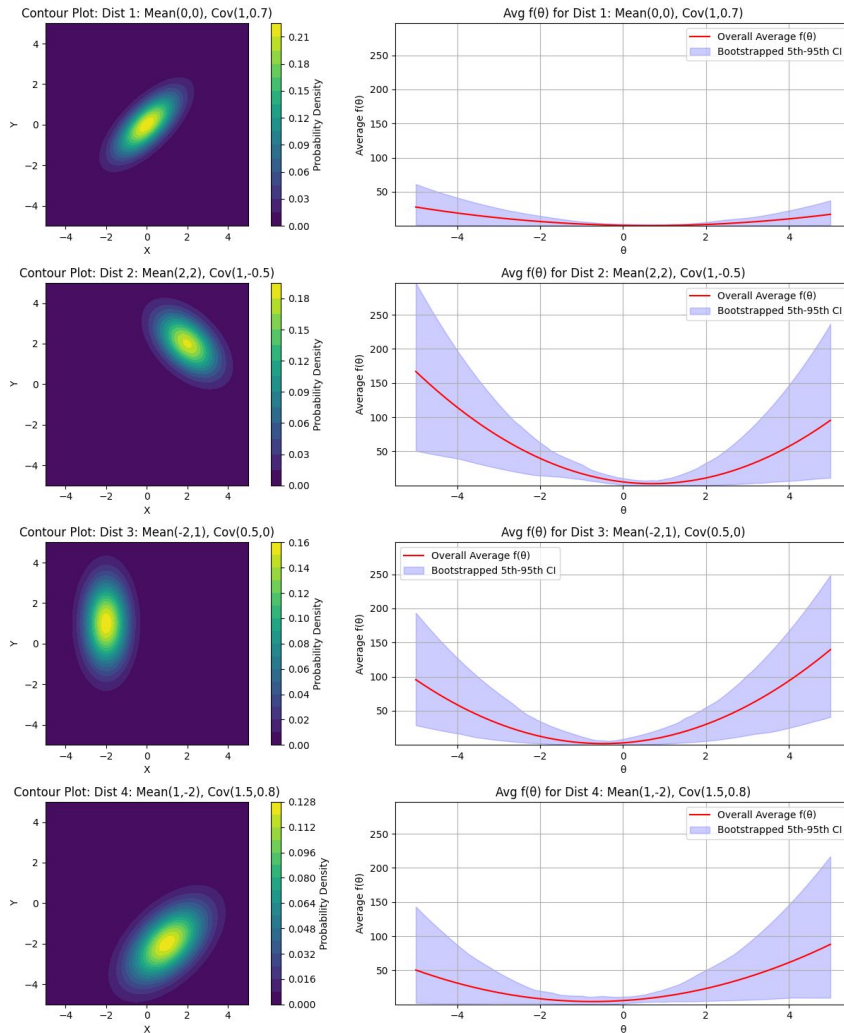
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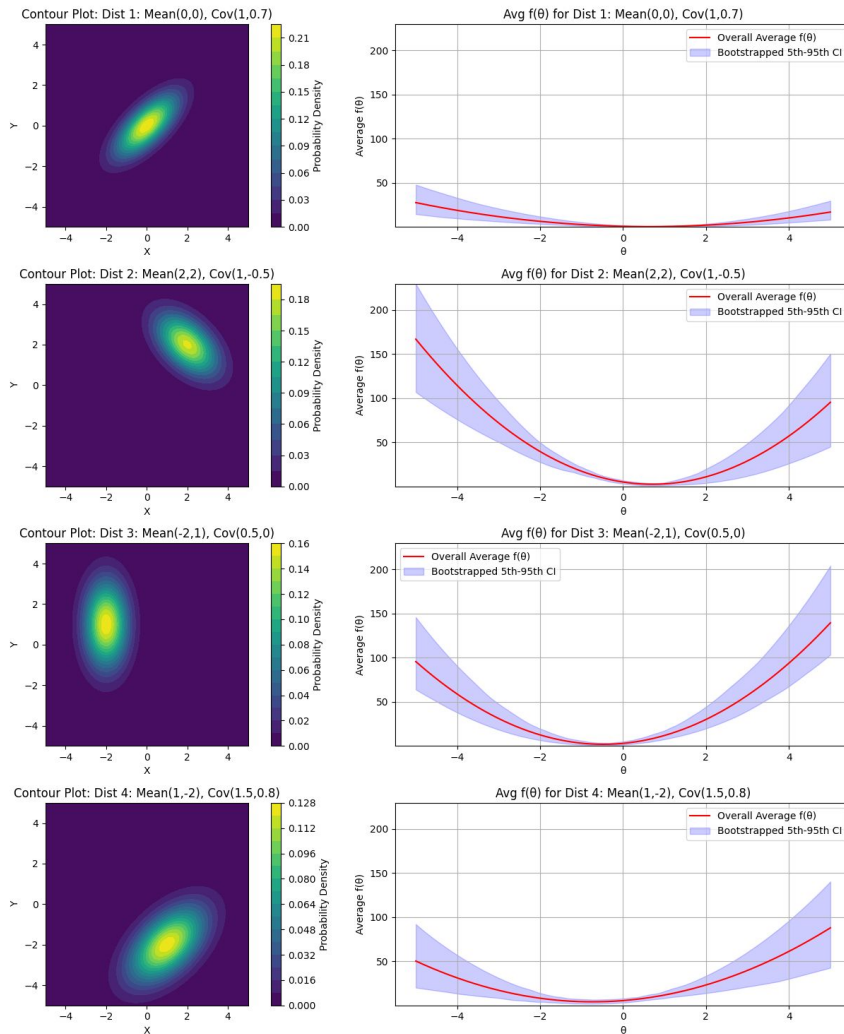
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What about more
complex probabilistic
objectives?

Probabilistic Optimization: Marginalizing Latent Structure (Mixture of Experts)

1. The Setup: Unknown Regimes

$$\left[\nearrow \right] x \in \mathbb{R}^d \quad \left[\text{target} \right] y \in \mathbb{R}$$

$$\text{person icon} z \in \{1, \dots, K\}$$

The Model:

- **Experts** (Fixed):
If regime is z , $y \approx e_z(x)$
- **Gating Network** (Unknown θ):
 $P(z = i | x) = \pi_i(x; \theta)$
(softmax gate)



Data generated by different mechanisms (experts) depending on unobserved context.

2. Marginalization: Averaging Out Latent Uncertainty

True conditional distribution:

$$p(y | x) = \sum_{i=1}^K P(z = i | x) \cdot p(y | x, z = i)$$

Optimal Predictor (Expected Value):

$$\hat{y}(x) = \mathbb{E}[y | x] = \sum_{i=1}^K \pi_i(x; \theta) e_i(x)$$

Crucial Point: The individual discrete latent variable z has **disappeared**. Replaced by a smooth, weighted average across all possibilities.

3. The Smooth Optimization Problem

Minimize MSE of marginalized predictor:

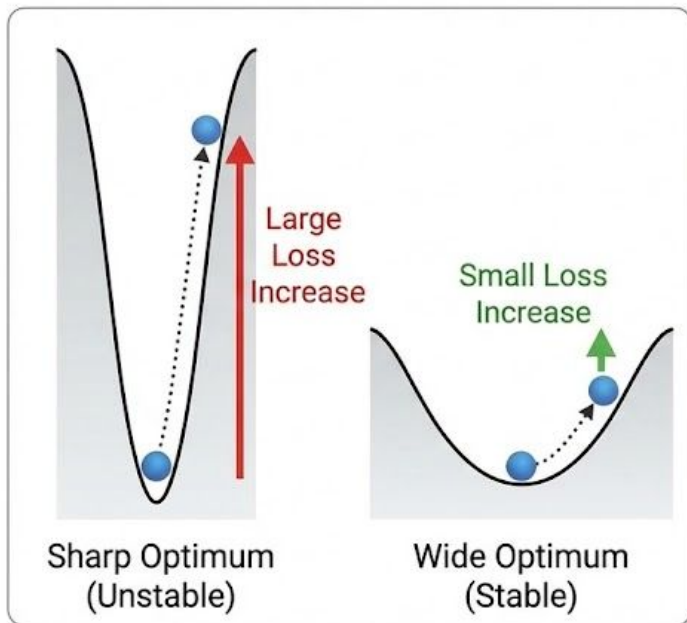
$$\min_{\theta} L(\theta) = \sum_{n=1}^N \left(y_n - \sum_{i=1}^K \pi_i(x_n; \theta) e_i(x_n) \right)^2$$

- Objective is smooth w.r.t. θ .
- Use standard Gradient Descent.

Concept Summary: The latent variable (z) exists in the probability model, but we integrate it out before defining the optimization objective. We never need to infer or optimize z directly. ✨

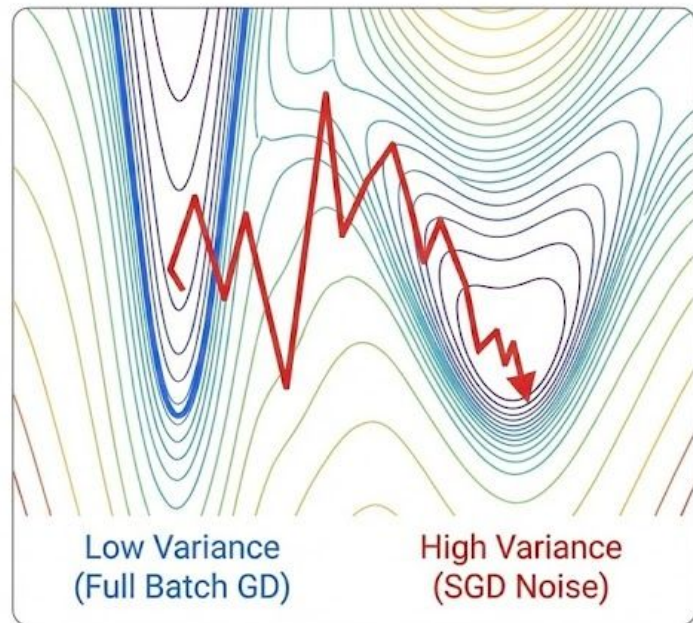
Probabilistic High Dimensionality

Sharp vs. Wide Optima & Stability



Small shifts in parameters cause massive loss changes in sharp optima.

Benefit of Gradient Variance (Noise)



Gradient noise helps escape sharp local minima and find flatter, more robust solutions.

Saddle Points: Unstable Stationary Points

A saddle point is a stationary point where the type of critical point differs across coordinates. It acts as a minimum in one direction and a maximum in an orthogonal direction, creating a characteristic saddle shape and representing an unstable equilibrium.

